

Problem Set 1

Given on: 13th September 2024

Due on: 27th September 2024

Total: **50 points + 4 bonus points**

Topics: Finite Automata & Regular Languages

Question 1

[20 points]

For each of the following languages L over the alphabet $\Sigma = \{0, 1\}$, give a regular expression that represents L and describe a DFA that recognizes it.

- (a) All strings of length divisible by 3 such that '10' is a substring.
- (b) All strings that contain '101' as a substring followed by an even number of 1's.

Question 2

[10 points]

For each of the following languages L over the alphabet $\Sigma = \{0, 1\}$, construct an NFA with the specified number of states recognizing L .

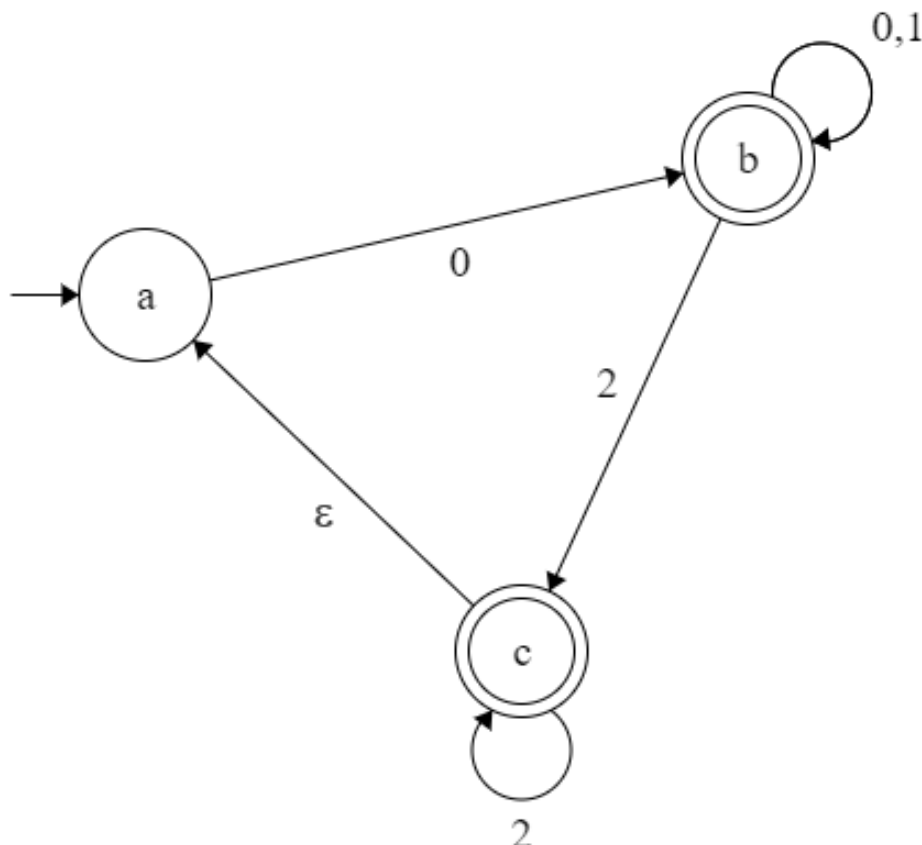
- (a) All strings have an odd number of 0's or 1's in multiples of 3, in six states.
- (b) All strings are in the form $1^*0^*1^+$ with three states.

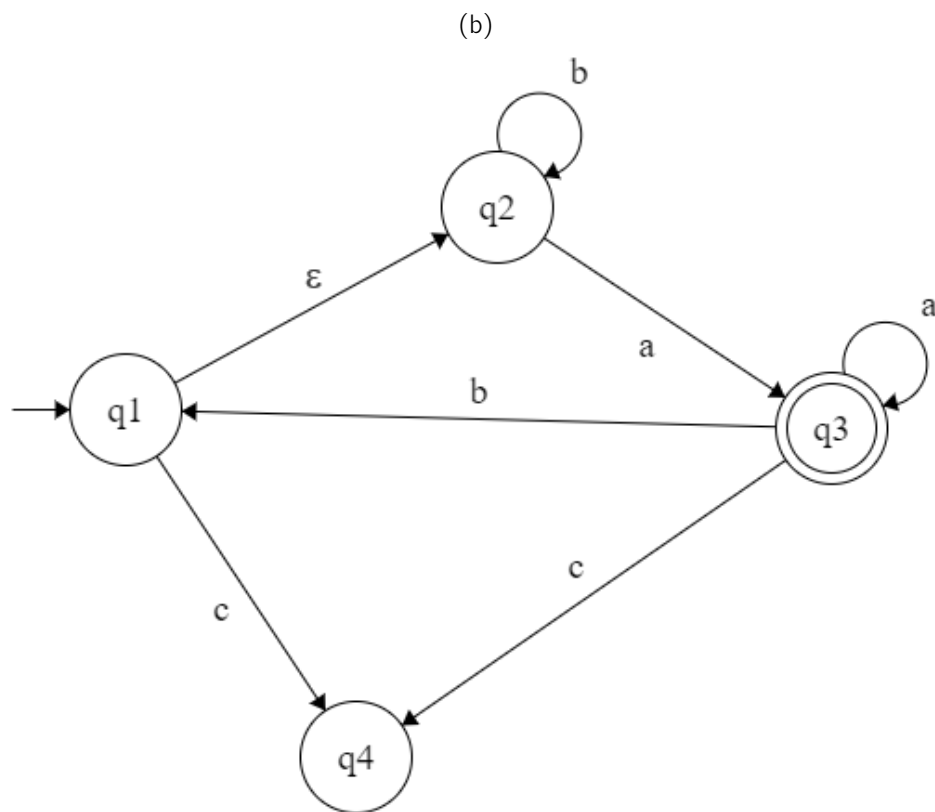
Question 3

[10 + 4 points]

Convert the following NFA's to equivalent DFA's. *Bonus*: find a regular expression for them.

(a)



**Question 4****[10 points]**

The suffix of a language L denoted $SUFFIX(L)$ is a set of all suffixes of strings in L . For example, if $L = \{ab, abc, b\}$, then $SUFFIX(L) = \{ab, b, bc, c, \epsilon\}$.

Show that if L is a regular language, then $SUFFIX(L)$ is also regular by constructing a new automaton that accepts all suffixes of L . Outline the process of modifying the DFA to recognize $SUFFIX(L)$.